Chroma 19501-K Tester PD Verification Procedure

Purpose: High-voltage production testing requires this <u>two-stage</u> procedure to be performed anytime the Chroma 19501-K Tester calls for a PD Verification Check. The PD Verification Check will be required anytime one of the following three conditions occur: (1) a Chroma 19501-K Tester is connected to a different GT Tester, (2) the OPN changes from the last run lot, or (3) more than 24 hours have passed since the last PD Verification Check. <u>Both stages</u> of this procedure must pass for the Chroma 19501-K Tester to be in a calibrated state and able to test production devices.

Hardware Required:

- **A195001 Verification Kit (see Fig. 1)**

Figure 1

- **BNC to Alligator Clip Cable (see Fig. 2)**

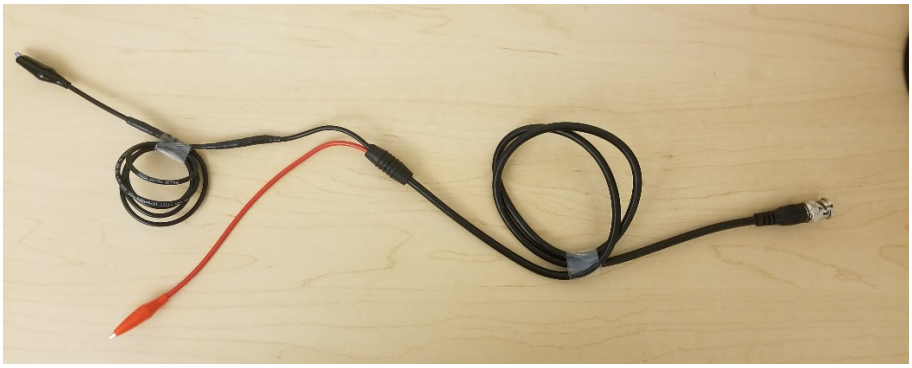

Figure 2

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Procedure:

1. Start a gtt7 session ('gtt7' in terminal), fill out the lot information for the devices to be tested, select the **"PD Verification"** flow, and click **Start Lot**

   (NOTE: NO HAZERDOUS VOLTAGES SHOULD BE PRESENT AT ANY TIME DURING THE PD VERIFICATION CHECK)

2. Connect the **A195001 Verification Kit** to the high-voltage setup as follows:
   a. Connect the **BNC side of the Cable** to the **200pC** port of the **A195001 Verification Kit** (see Fig. 3)

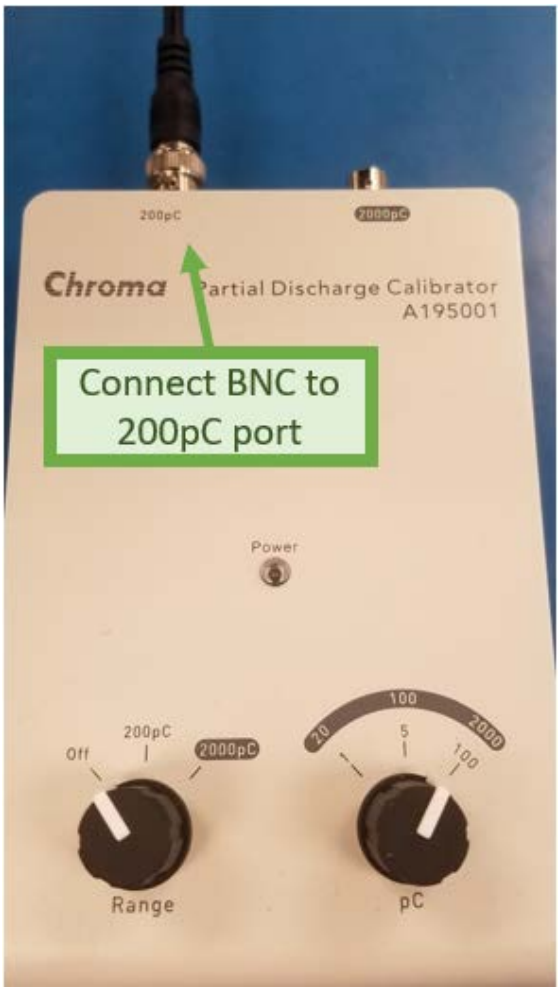

Figure 3

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b. Open the **Plexiglass Cover** on the handler and **remove the Contact Site Cover**. Connect the **Red Alligator Clip side of the Cable** to the **RIGHT-SIDE contact plate of the handler** (see Fig. 4)
(NOTE: The PD Verification Check will NOT pass if the connection is made to the LEFT-SIDE contact plate of the handler)

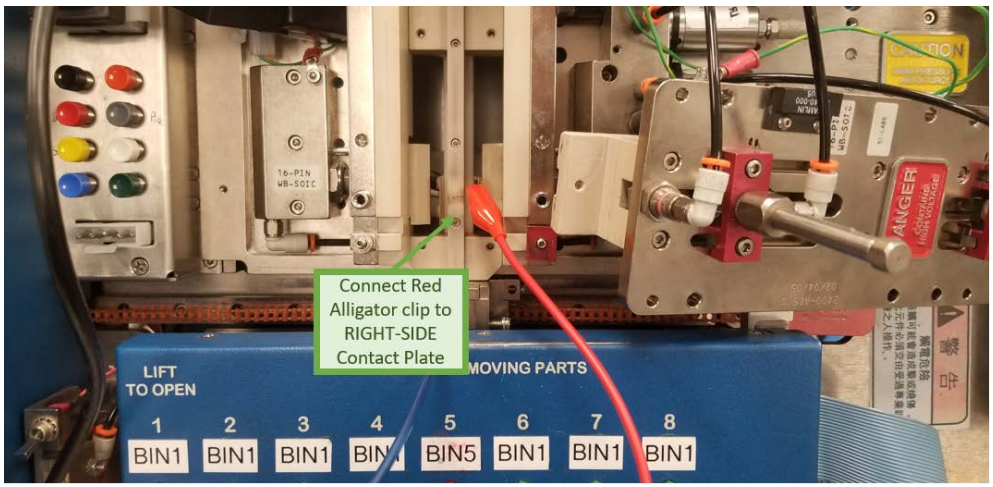

Figure 4

c. Connect the **Black Alligator Clip side of the Cable** to the **Chroma 19501-K Test Head Chassis Ground** (see Fig. 5)

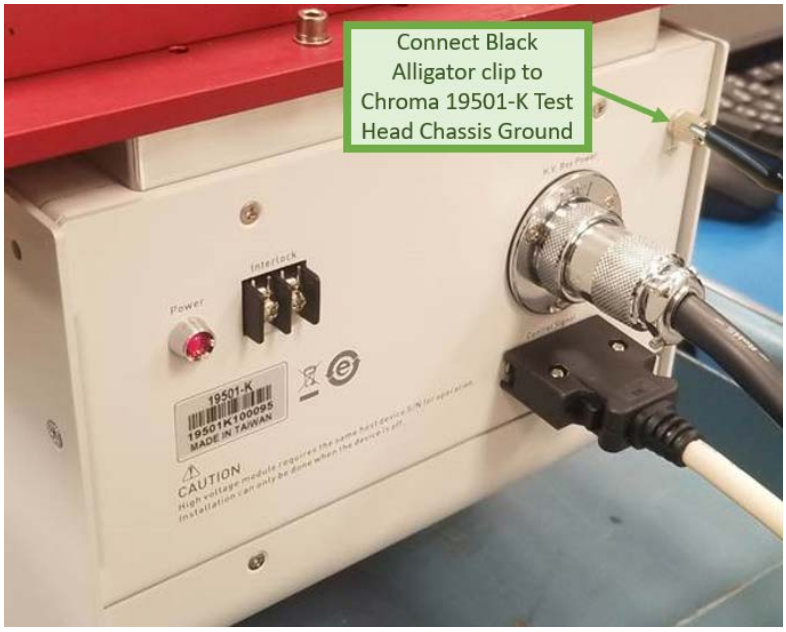

Figure 5

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d. Carefully close the **Handler Plexiglass Cover** making sure the **BNC to Alligator Clip Cable** routes under the **bottom of the handler** and does not become crimped or damaged when the **Handler Plexiglass Cover** is closed (see Figs. 6 & 7)
(NOTE: The PD Verification Check will not start if the Handler Plexiglass Cover is open)

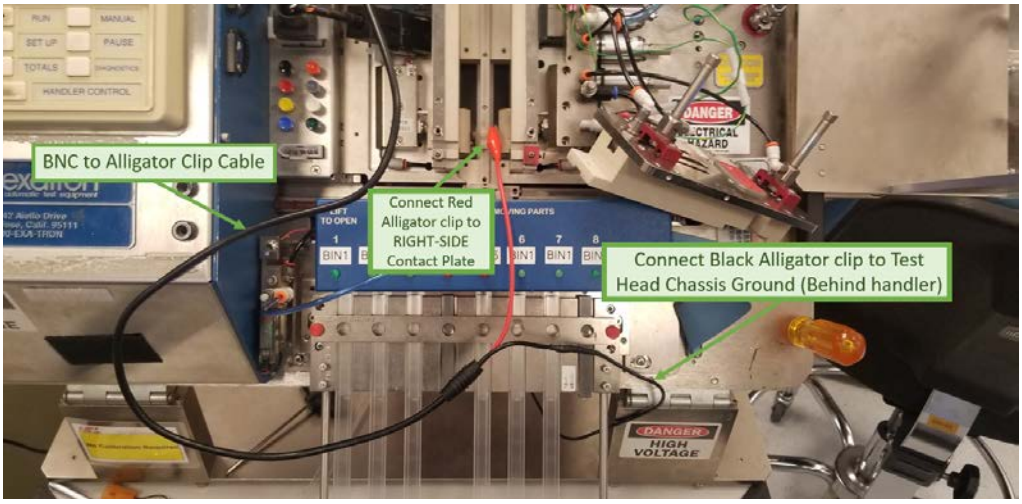

Figure 6

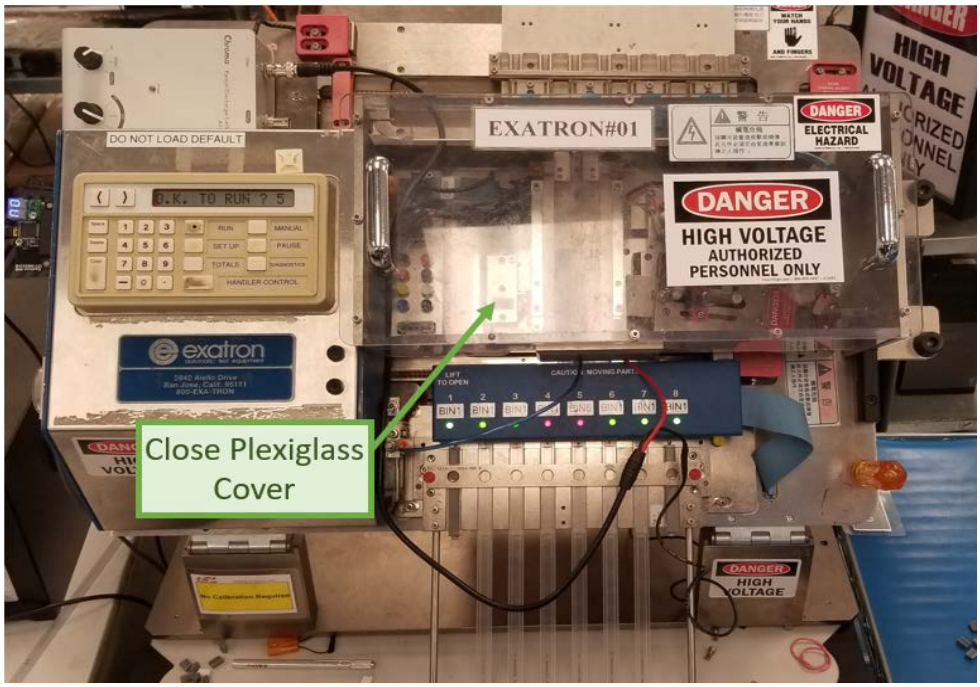

Figure 7

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STAGE 1

3. Set the **Range** dial on the **A195001 Verification Kit** to **200pC** and the **pC** dial to 1 (See Fig. 8). (NOTE: The green Power LED should now be enabled on the A195001 Verification Kit)

Figure 8

4. Click **Start Test** on the GT GUI. Another GUI will pop-up with the previous instructions outlined in steps 1-3. Ensure that all the instructions have been correctly followed and click **OK.**

5. The Chroma 19501-K Tester will now begin the PD Verification Check for **Stage 1**. This process will take **around 5.0 seconds to complete**. For **Stage 1**, the Chroma 19501-K Tester will occasionally be **unable to get an accurate PD reading and the PD Verification Check will fail**. If this occurs, click **Start Test** to re-try (this may take up to 5 more times to get an accurate reading). If the procedure is still unable to pass, please contact Engineering.
(NOTE: The **RED DANGER LED** will blink while the PD Verification Check is occurring)

6. If the results of **Stage 1** are successful, another GUI will pop-up with instructions for **Stage 2** of the PD Verification Check.

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STAGE 2

7. Leave the **A195001 Verification Kit** connected and change the **pC** dial to **100** (See Fig. 9).

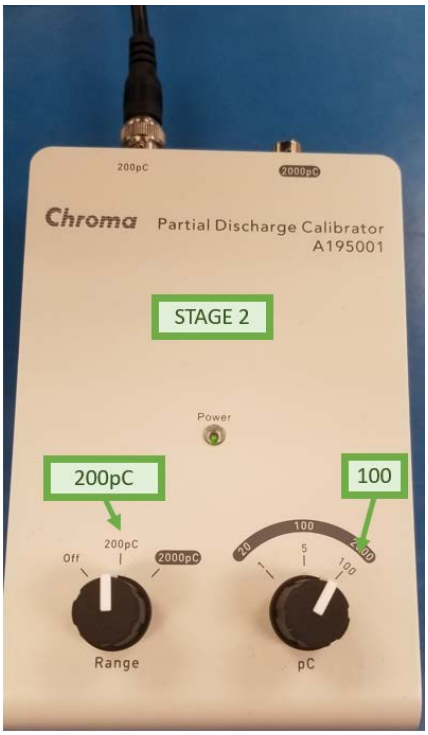

Figure 9

8. Click **OK** in the GUI pop-up.

9. The Chroma 19501-K Tester will now begin the PD Verification Check for **Stage 2**. This process will take **around 5.0 seconds to complete**. If the procedure is unable to pass, please contact Engineering.
   (NOTE: The **RED DANGER LED** will blink while the PD Verification Check is occurring)

10. If the results of **Stage 2** are successful, click **End Lot** in the GT GUI, change the **Range** of the A195001 Verification Kit to **OFF**, and disconnect the A195001 Verification Kit from the setup.

11. Exit the GT GUI and start a new gtt7 session ('gtt7' in the terminal). This time select the **"Production"** flow when filling out the Lot Information to begin normal testing.

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